

Article

Argo Buoy Trajectory Prediction: Multi-Scale Ocean Driving Factors and Time–Space Attention Mechanism

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Abstract: The Array for Real-time Geostrophic Oceanography (Argo) program provides valuable data for maritime research and rescue operations. This paper is based on Argo historical and satellite observations, and inverted sea surface and submarine drift trajectories. A neural network method was developed to predict the position of Argo buoys, improving target tracking and emergency support capabilities. Based on a deep learning framework using a Simple Recurrent Unit (SRU), a new Time–Space Feature Fusion Method based on an Attention Mechanism (TSFFAM) model was constructed. The TSFFAM mechanism can predict the target trajectory more accurately, avoiding the disadvantages of traditional Long Short-Term Memory (LSTM) models, which are time consuming and difficult to train. The TSFFAM model is able to better capture multi-scale ocean factors, leading to more accurate and efficient buoy trajectory predictions. In addition, it aims to shed light on the mechanism of the joint multi-element and multi-scale effects of laminar and surface currents on multi-scale ocean factors, thereby deepening our understanding of the multi-element and multi-scale interactions in different spatio-temporal regimes of the ocean. Experimental verification was conducted in the Pacific Ocean using buoy trajectory data, and the experimental results showed that the buoy trajectory prediction models proposed in this paper can achieve high prediction accuracy, with the TSFFAM model improving the accuracy rate by approximately 20%. This research holds significant practical value for the field of maritime studies, precise rescue operations, and efficient target tracking.

Keywords: Argo; neural network; deep learning; TSFFAM; target tracking



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1. Introduction

Ocean phenomena are continuous, complex, and ever-changing in both space and time. The Array for Real-time Geostrophic Oceanography (Argo) program deploys a satellite tracking buoy every 300 km worldwide. The EUROARGO system uses new, innovative sensors to collect many temperature, salinity, and dissolved oxygen profiles [1]. The floats are generally configured to slightly ascend and continue drifting in case of grounding, achieving a continuous drift and a better representation of water movement, even in shallow areas. The typical life cycle of an Argo float is shown in Figure 1 below. The implementation of the Argo program aims to enhance the precision of climate forecasting, thereby mitigating the escalating hazards of global climatic disturbances that pose a significant threat to human society [2,3]. The existing trajectory tracking prediction methods can be mainly divided into two categories: statistical approaches that require feature extraction, feature engineering, and algorithm selection, and deep learning-based approaches that only require algorithm selection and optimization. Traditional statistical data-driven life prediction methods assume that the degradation model is known in advance and use monitoring data to estimate the parameters of the model online or offline.

However, in practical applications, the degradation model is unknown, and improper model selection has a significant impact on prediction accuracy [4,5]. The early research on trajectory prediction mainly focused on simulating the trajectory of moving objects [6,7]. The above-mentioned algorithms have achieved good prediction results, but only predict discrete position points and not continuous position information such as longitude and latitude coordinates. The long-term and short-term memory network—usually referred to as “LSTM”—is a special RNN that can learn long-term rules. It was first proposed by Hochreiter (1997) and refined and popularized by many people later. With the rapid development of deep learning, Long Short-Term Memory (LSTM) [8–11] is also widely used in trajectory sequence prediction tasks. The prediction methods based on deep learning avoid unknown degradation, extract effective information from monitoring data, depict the nonlinear relationship between feature information and life, and can track and predict the trajectory more accurately [12–14].

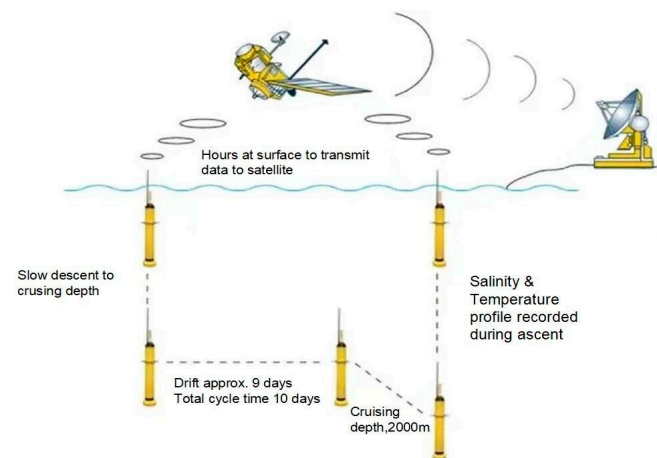


Figure 1. The typical cycle of an Argo float [15]. The buoy begins to descend to 1000 m above the sea surface and stays at that depth for 9–10 days. Typically, after 9 days, the buoy will automatically descend to a depth of 2000 m, activate its own sampling sensor, and measure marine environmental factors as it rises to the surface. When the buoy reaches the sea surface, it will drift on the surface and transmit the collected data to the satellite system. Subsequently, the buoy descends again to the drift depth and begins the next cycle. The entire cycle is usually 10 days.

In summary, due to the complexity of trajectory prediction, the existing intelligent predictions of the ocean mainly focus on aspects such as seawater environment and marine disasters [8,9,11,12,14,16–25]. Currently, Bayesian model-based methods are not applied in trajectory research [26]. Instead, deep learning methods, such as transformers, are more commonly used in traffic-related predictions including pedestrians and vehicles [27–33]. This paper analyzes the structural characteristics and spatio-temporal distribution traits of Argo data, and combines spatio-temporal feature fusion technology to achieve the explicit conversion of irregular and discrete Argo profile data into continuous grid data. When they are working, the trajectory of Argo buoys is diversely complex at different time and spatial scales. This is not only related to the drifting trajectory on the sea surface, but is also closely related to that below the sea surface. Therefore, it is necessary to comprehensively analyze and predict the trajectory of Argo buoys by combining both the surface- and middle-layer current elements. Consequently, this paper designs a module that combines multi-scale temporal and spatial attention mechanisms to extract the features from sea surface and underwater spatio-temporal trajectories, respectively. Then, the multi-scale temporal and spatial features are integrated, and finally the trajectory of the Argo buoys using different models is analyzed and predicted.

The Simple Recurrent Unit (SRU) first proposed by Lei (2018) [34] is a light recurrent unit that balances model capacity and scalability. The SRU is designed to provide expressive recurrence and enable highly parallelized implementation, and can facilitate the training

of deep models. Based on the deep learning framework of the SRU, a TSFFAM model was constructed that utilizes an attention mechanism to predict buoy information [35–42], effectively avoiding the problems of traditional LSTM models such as time-consuming prediction and difficulty in training. Due to the limitations of observational data, the understanding of the 3D Pacific circulation in the oceanographic community has long focused on surface circulation, while significant knowledge gaps exist regarding the structure, variability, and links to climate change of flows in the deep subsurface mesosphere. The TSFFAM model is able to better capture multi-scale ocean factors, leading to more accurate and efficient buoy trajectory predictions. The spatio-temporal attention mechanism can be used to achieve more accurate and efficient prediction of buoy trajectories, thereby revealing the multi-element and multi-scale joint action mechanism of laminar and surface currents in multi-scale ocean factors, and deepening the understanding and recognition of multi-element and multi-scale interactions in different spatio-temporal environments in the ocean.

2. Materials

The dataset used in this paper was downloaded from Argo GDAC (Global Data Assembly Center <https://argo.ucsd.edu> accessed on 31 December 2023.) with a time range from July 1997 to September 2023. The satellite observations were sourced from the altimetry data retrieved from the Archiving Validation and International of Satellites Oceanographic (AVISO) website (<http://marine.copernicus.eu/> accessed on 24 October 2023). The data span from 1997 to 2023. The surface flow data were calculated by inversion based on the trajectory of the buoy (Figure 2).

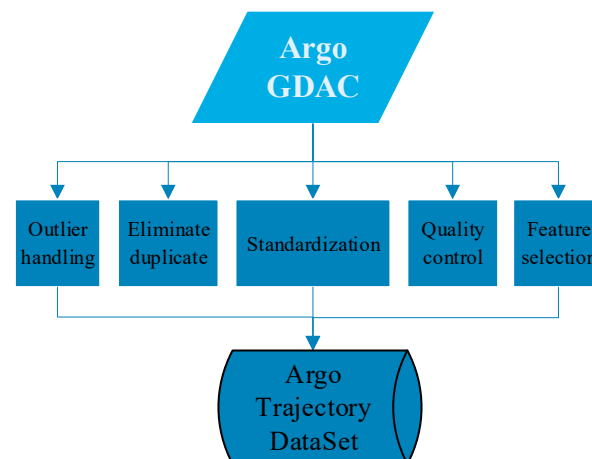


Figure 2. Flowchart of the data process. The data processing procedure commences with the removal of outliers, followed by deduplication and standardization of the data. Subsequently, historical data are integrated for quality control purposes, culminating in the generation of a feature-based dataset.

During the production of the dataset, buoy data quality re-control is carried out, including 16 detection steps, such as observation time, longitude and latitude, and buoy offset velocity detection [43–55]. Buoy drift velocity detection considers the longitude, latitude, and time of the current and previous sections and marks if the speed is greater than 2.0 m/s [56–65].

(1) Outlier handling

Outlier analysis was performed to check whether the data have been entered incorrectly or unreasonably. This included checking whether the current and previous record types of each line in the data file are legal, and examining the arrangement order, starting position, length, data storage type, character code, etc. First of all, for some buoy samples with obvious anomalies or errors in the quality control results, it was necessary to locate the position of these bad data. The causes of errors in the bad data were analyzed and marked.

Then, the machine learning algorithm could be used to pre-experiment on the buoy data. The Argo buoys are cycled every 10 days for data transmission. Individual Argo buoys are designed by the experimenter to cycle for less than 10 days and are not included in the Argo trajectory prediction.

(2) Eliminate duplicates

This began by establishing the parameters (time, location, observation values, etc.) necessary for identifying the duplicated data. Next, the time and location of the relevant data were sorted and compared, deleting any identical duplicates. The suspected duplicate data were preserved if they originated from more reliable sources (with more complete and accurate information), while the data from the other sources were labeled as being duplicative and manually eliminated.

(3) Data standardization

Data standardization is the process of scaling data proportionally to fit into a smaller specific interval $[-1, 1]$, removing unit limitations from the data, and converting them into dimensionless pure numerical values for comparing and weighing the indicators for different units or magnitudes. For longitude data, the difference between 179° W and 179° E is only 2° in geographical location, but it is 358° as a numerical value, thereby greatly increasing the difficulty of model learning. In order to express the continuity of the original longitude and latitude information in numerical terms and the connectivity between the beginning and end, the longitude and latitude information were encoded separately to expand the data dimension, effectively improving the accuracy of model prediction.

(4) Quality control

The methods and steps of quality control of ocean profile data are used to automatically control the quality of profile data. This mainly includes scope inspection, gradient inspection, climatology inspection, and other quality control methods and parameters, and the manual review of the quality control results.

(5) Feature selection

Feature selection needs to analyze the importance of each feature in the original dataset. The features of less importance were removed to reduce the difficulty of model learning and to avoid dimensionality. Due to the fact that machine learning methods do not accept inputs having string types, all string type data were converted using one-hot encoding.

3. Methods

In this study, LSTM and SRU models were built and tested. Simultaneously, a comparative experiment was conducted by adding a spatio-temporal mechanism module to the two models. The SRU's parallel computing capabilities make it more efficient than the traditional LSTM techniques. This model adopts a novel methodology to tackle the concerns regarding computation time and complexity. A skip connection strategy was implemented to address overfitting and the vanishing gradient issues, and the SRU improvement strategy was used to significantly improve the predictive accuracy during model optimization. The following is a brief introduction to the models used in this paper.

3.1. LSTM

The structure of the LSTM unit is illustrated in Figure 3. The input gate at time step t has the ability to select and input the information from the current step t to produce the result shown in Equation (1). The forgetting gate selectively forgets the hidden layer output in the previous step (Equation (2)) and combines it with the memory of the previous step to generate the memory of the current step (Equation (3)). The output gate is responsible for determining the final output of that step (Equation (4)).

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (1)$$

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

$$c_t = f_t \otimes c_{t-1} + i_t \otimes \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (3)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (4)$$

$$h_t = o_t \otimes \tanh(c_t) \quad (5)$$

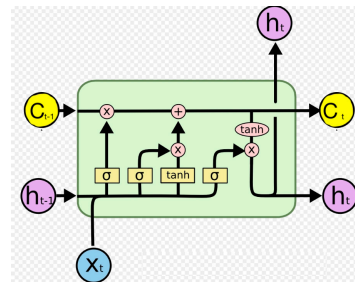


Figure 3. Network structure of LSTM cell. The input values of all structures are cleverly integrated by point multiplication and addition operations, and are updated in iterative training of the model. Finally, the activation function controls the output values between 0 and 1, where 0 represents no output and 1 represents all outputs (Equation (5)).

3.2. SRU

The input gate and output gate are used to receive, output, and correct parameters, which are denoted by i and h . The forgetting gate is denoted by f , indicating the historical information stored by the current hidden layer node. The memory unit (cell) is denoted by c , which represents the memory of the neuronal states. The design of the gated cell unit enables the SRU unit to save, read, reset, and update long-distance historical information. The mechanism of the SRU network structure is as follows (Figure 4):

$$\tilde{x}_t = Wx_t \quad (6)$$

$$f_t = \sigma(W_f x_t + b_f) \quad (7)$$

$$r_t = \sigma(W_r x_t + b_r) \quad (8)$$

$$c_t = f_t \odot c_{t-1} + (1 - f_t) \odot \tilde{x}_t \quad (9)$$

$$h_t = r_t \odot g(c_t) + (1 - r_t) \odot x_t \quad (10)$$

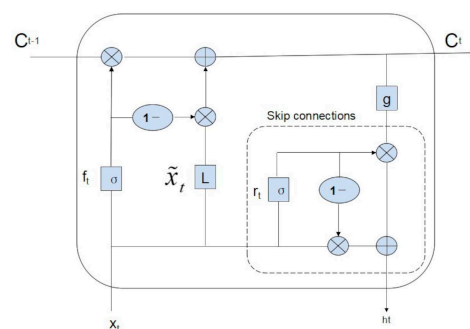


Figure 4. Network structure of SRU cell, where f and r represent the forget gate and the reset gate, respectively. h and c are the output state and the internal state; x represents the input; W and b are the corresponding weight coefficient matrix and bias terms, respectively; σ is the activating sigmoid function.

The SRU does not rely on the output of the previous unit, thus achieving the parallel processing of each unit and improving the model training speed. Another advantage is

that the SRU has fewer parameters than the LSTM, making the model easier to converge. This paper suggests an SRU and attention model-based network structure to resolve this issue. Most of the operations are processed in parallel, but there are some steps with serialized operations. In order to offset the significant increase in computational complexity, parallelization methods such as GPU accelerated training have been widely accepted to scale deep learning.

3.3. Model Establishment

This study used a three-layer deep learning network based on LSTM and SRU, and two independent neural network models were built for comparative testing. The SRU model was built in parallel with three layers of SRUs. Each layer had 256 neurons, with a dropout rate of 0.2 and a learning rate of 0.001~0.1. The multi-year climatic current was used as the background and then the input parameter for the more reliable prediction of the trajectory. The buoy trajectory data and seasonal characteristics of the current were integrated as the model hyperparameter inputs.

This discovery realizes the construction of a multi-layer SRU. Because the activation function and the sigmoid function need their own independent functions, it adds an additional running delay and data movement overhead. Therefore, this invention is based on technologies such as grid and parallel computing, and optimizes and upgrades the multi-layer SRU model, significantly improving its computational efficiency and GPU parallel computing performance. The specific steps are as follows:

For neural networks with unequal dimensions, three weight matrices of the neural network are merged into a large matrix through linear transformation.

Element-by-element multiplication into a kernel function allows for the parallel processing of matrix multiplication at all of the time steps, significantly improving the computational efficiency and GPU utilization.

Through operations such as gridding and parallel computing, the outer two “fors” in the kernel function can achieve parallel operations. The innermost “for” has a sequence order, and only this dimension needs to be correlated before and after. Parallel computing can be separated into both the “minibatch” and “hidden State” dimensions. Finally, the sequence dimension that needs to be managed before and after can be placed in a register to maintain a sequential relationship. The dimensions of the minibatch and hidden State represent the x and y axes of the grid, respectively.

3.4. Model Optimization

An attention mechanism can also reduce the aliasing effect caused by feature fusion. In order to improve the detection accuracy of trajectory features, in this paper, a Time-Space Feature Fusion Method based on an Attention Mechanism (TSFFAM) was first designed, which not only suppresses sequence-useless information in time, but also suppresses the weight-irrelevant information in space.

Due to the residual network structure being able to reduce the loss of feature information during the feature extraction process, a skip residual connection module was designed to reduce the loss of high-level feature information during the feature fusion process. Finally, based on the advantage of the capability to extract features of different depths, the TSFFAM jointly learns the attention weights of different channels in the spatial dimension and those of the different frames in the temporal dimension. The features X_i and $A_j \in \mathbb{R}^{H \times W \times T \times C}$ are passed into these two modules in a sequence. The TSFFAM sequentially infers spatial $M_s \in \mathbb{R}^{1 \times 1 \times 1 \times C}$ and temporal attention maps $M_t \in \mathbb{R}^{1 \times 1 \times 1 \times T}$.

$$\begin{aligned}
 X_i' &= M_s (X_i) \otimes X_i \\
 Y_i &= T_{\text{trans}} (M_s (X_i') \otimes X_i') \\
 A_j' &= M_t (A_j) \otimes A_j \\
 B_j &= T_{\text{trans}} (M_t (A_j') A_j') \\
 L_k &= Y_i + B_j
 \end{aligned} \tag{11}$$

The symbol $\otimes$ represents element-wise multiplication. The final refined output, denoted as L_k , is then passed into the next module in the model.

In addition to the spatial information, temporal information is also crucial for trajectory prediction. It is challenging to infer the speed of a moving buoy without time data. To solve this problem, building precise associations between the features at different times is a challenge. Therefore, the TSFFAM module modeled this spatio-temporal correlation, extracting temporal information features from the previous data instead of individual stacked data features. Based on the TSFFAM module (Figure 5), long-term dependency relationships can be more effectively modeled, requiring a smaller computational cost and reducing the quantity of incorrect information.

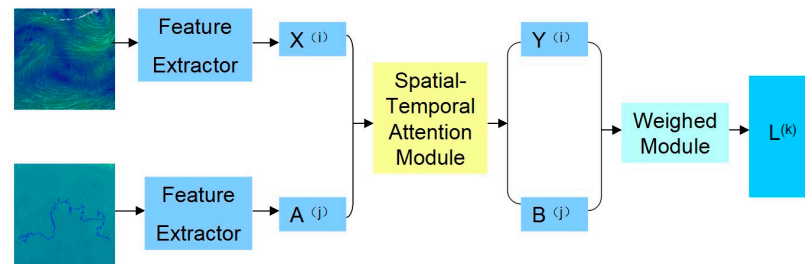


Figure 5. Model diagram. X represents the trajectory sequence formed by statistical analysis of multi-source data, while A represents the current input sequence of buoy trajectories. The extracted multi-scale temporal and spatial features are fused to carry out an analysis and prediction of Argo buoy trajectories using optimal model weights.

3.5. Model Evaluation

The core goal of establishing a regression model is to minimize the sum of the squared differences between predicted and true values on the training set as much as possible. The calculation formula is:

$$\sum_{i=1}^m \left(y_{\text{train}}^{(i)} - \hat{y}_{\text{train}}^{(i)} \right)^2 = \sum_{i=1}^m \left(y_{\text{train}}^{(i)} - \alpha x_{\text{train}}^{(i)} - b \right)^2 \quad (12)$$

The symbol α represents weight and b represents deviation. After model training and optimization, the optimal parameter configuration was selected to make the predicted value of the model closest to the true value.

In order to evaluate the predictive ability of the model, it is necessary to establish intuitive quantitative standards for measurement. Due to the unique nature of longitude information, two points with a geographical location difference of 2° can have a numerical difference of up to 358° . The longitude and latitude of the buoy are predicted separately for the sake of overall model accuracy. During the training phase of this study, the model's training objective is to calculate the distance between the actual longitude/latitude value and the predicted value. The formula used for model evaluation does not identify these predicted results. The mean square error (MSE) is selected as the loss function, the back propagation algorithm is used to calculate the gradient of each weight, the gradient optimization algorithm is used to update the weight, and the buoy positions in the training set are trained in batches until convergence. In order to eliminate the influence of dimensionality, the root mean square error (RMSE) is calculated from the square of the mean square error (MSE) as a second evaluation metric. The reason for using RMSE instead of the information bound is that the RMSE balances biased and unbiased estimations when evaluating our model [66,67]. This measures the size of the difference between the predicted and true values. The smaller the value, the smaller the difference between the result and the true value.

Comparing the effectiveness of models under different dimensions requires the use of the regression model's evaluation index: the R -squared value (R^2). $R^2 = 0$ indicates that every predicted value in a sample is equal to the average value. $R^2 = 1$ denotes that the

predicted and true values are identical, with no error. As the errors increase, R^2 diminishes, moving farther from the maximum value of 1.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n [\hat{y}(i) - y(i)]^2} \quad (13)$$

$$R^2 = 1 - \frac{\sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2 / m}{\sum_{i=1}^m (y^{(i)} - \bar{y})^2 / m} = 1 - \frac{MSE(\hat{y}, y)}{Var(y)} \quad (14)$$

4. Results

4.1. Experiment Setting

The Argo dataset has a time range from 1997 to 2023. The training dataset includes 6000 buoy trajectories from 1997 to 2020, and 1000 buoy trajectory values from 2021 to 2023 represent the validation dataset. Then, the LSTM, SRU, LSTM with TSFFAM, and SRU with TSFFAM models were applied for training. Furthermore, the precision and effectiveness of our spatio-temporal attention module were validated. The time sequences were shifted at ten-day intervals, thus creating secondary, tertiary, quaternary, and subsequent paths. The first ten-step outcomes of each sequence were fed as inputs into the model, whereas the output of the eleventh step was employed to assess the prognostic accuracy of the model. The complete trajectory sequence of all the buoys constitutes dataset D. Due to the temporal characteristics of the trajectory data, in order to avoid future information leakage caused by random partitioning, the trajectory sequence of each buoy sample was divided along the time axis into units of 365 days per year, resulting in training set S and testing set T. In order to avoid gradient descent falling into the local minima and to improve computational efficiency, the data in the training set were input into the model in batches for training. In the training process, some units and their connections were randomly discarded; in other words, the dropout mechanism was introduced to prevent the model from overfitting. The hyperparameter combination with the minimum loss function was finally selected for the model as shown in the Table 1 below. Once the hyperparameter configuration was determined, the model was retrained using the complete training set to obtain the final trajectory prediction model. The test set was then used to evaluate the model's predictive ability on data outside of the training samples.

Table 1. Hyperparameters of prediction model.

Parameter	Value
Learning rate	0.001~0.1
Batch Size	8~128
Epoch	30
Dropout	0.2

4.2. Experiment Results

The results of the experiment are presented in Table 2 and Figure 6. The SRU with TSFFAM model achieved the highest accuracy in the shortest time, which was only slightly higher than that of the SRU model. While the LSTM model had the lowest accuracy, the LSTM with TSFFAM model took the longest time, which further demonstrates the obvious advantages of the SRU with TSFFAM model.

Table 2. Comparison of LSTM and SRU prediction results.

Batch	Model	R^2	RMSE	Params	FLOPs
128	LSTM	0.56	4.5	24,503,255	127,019,530
128	SRU	0.68	3.2	12,234,268	73,405,608
8	LSTM + TSFFAM	0.76	1.7	233,445,676	1,400,674,056
8	SRU + TSFFAM	0.87	1.2	12,323,498	67,394,098

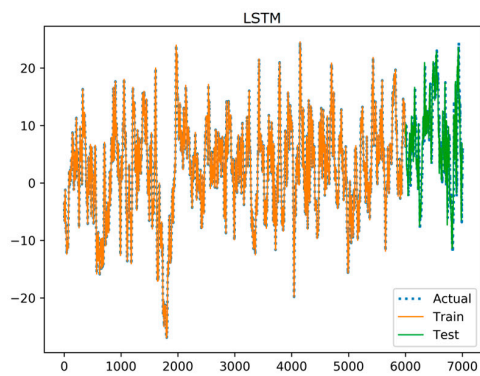
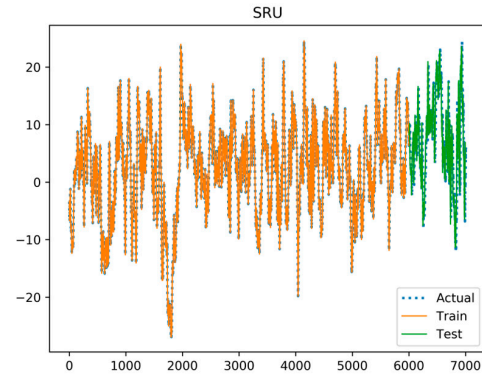
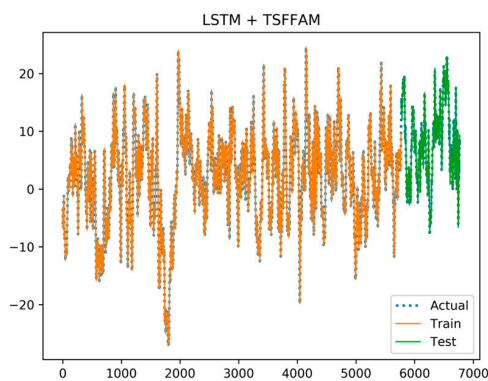
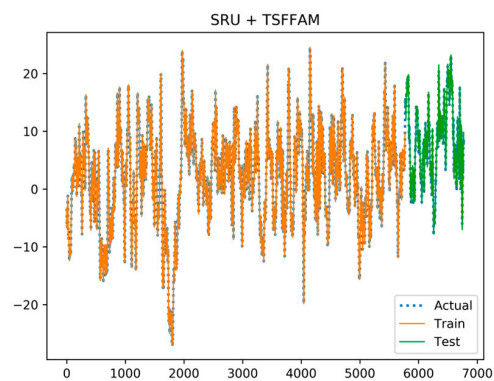
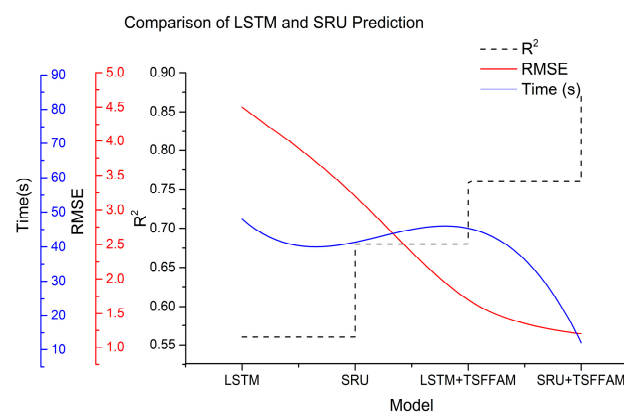
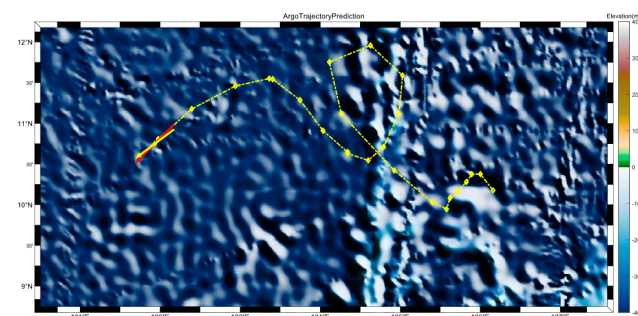
**(a)** LSTM Prediction Results**(b)** SRU Prediction Results**(c)** LSTM with TSFFAM Prediction Results**(d)** SRU with TSFFAM Prediction Results**(e)** Comparison of LSTM and SRU Prediction Results

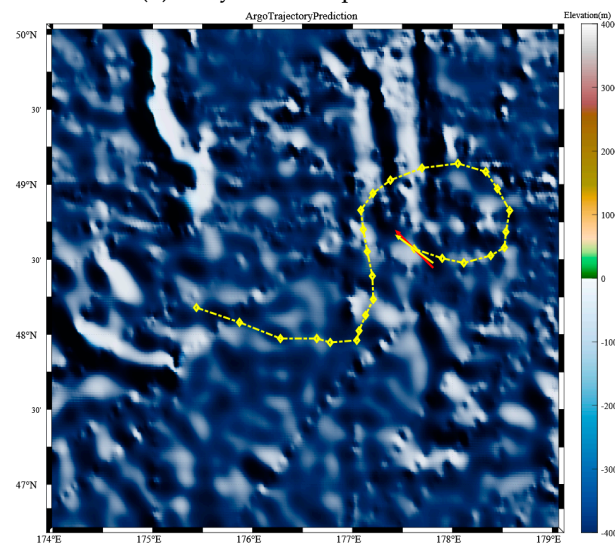
Figure 6. LSTM and SRU prediction results. **(a,b)** display the prediction results of the LSTM and SRU models, respectively. **(c,d)** show the prediction results of the LSTM and SRU models with TSFFAM, respectively. **(e)** shows the comparison of the prediction results and efficiency of prediction. The graph displays the advantage of the attention mechanism in TSFFAM methods, as indicated by the dashed lines, resulting in high goodness of fit values.

4.3. Experiment Validation

A validation experiment was carried out to test the accuracy of the predictive model with buoys 5,902,519 and 5,905,288 at 2 degrees of latitude (high and low) and at different longitudes. The result of the validation is shown in Figure 7. The true values are shown in yellow, and the red arrows represent the predicted values. It can be seen that the deviation of the prediction from the true value is extremely slight. This also demonstrates the validity of our proposed spatio-temporal attention module. Based on the analysis of the working principle of Argo buoys, they not only drift on the sea surface, but also drift underwater for up to 9 days. Accordingly, for the analysis and prediction of Argo buoy trajectories, simply using sea surface spatio-temporal trajectories for prediction is far from enough, and the integration of underwater values is needed. Through the experimental verification and analysis of the measured data, it is shown that using coupled spatio-temporal multi-scale process models and algorithms, integrating multiple elements, can effectively improve the accuracy of the analysis and prediction of Argo buoy trajectories.



(a) Buoy 5,902,519 predicts results



(b) Buoy 5,905,288 predicts results

Figure 7. Buoy (a) 5,902,519 and (b) 5,905,288 prediction results. The actual values are in yellow, while the red arrows indicate the anticipated values. It is noticeable that the difference between the projected value and the actual value is minimal.

5. Discussion

The movement of marine buoys is influenced by different environmental factors, leading to challenges in consistently predicting their paths, particularly when regional variances are considered. Due to the constantly shifting marine environment, this study's prediction window is limited, and predicting buoy positions over long periods is not practical. Due to various external factors affecting the accuracy of the trajectory data of Argo buoys, as well as significant differences in their drift patterns in different regions, the

prediction difficulty level of the model is relatively high. The prediction achieved using this model is expected and is a great reference for guiding target tracking and emergency support. In order to solve the error problem in track prediction, this study used natural language processing for attention mechanism prediction and an attention model for buoy tracking in a specific hidden layer by extracting the key features. The model focuses more on the key features of the input sequence, reducing the error caused by abnormal data in the prediction results. The results of the experiment are presented in Table 3.

Table 3. Accuracy of prediction model.

K	Sample		Accuracy		
	Train Set S	Test Set T	Lon	Lat	Avg
1	8760	26,280	82.90%	84.75%	83.82%
3	26,280	8760	86.03%	89.85%	87.94%

In this study, buoy trajectories with longer observation times were selected (no less than four years) as examples for training and prediction to display our prediction results. In order to evaluate the performance of our method, we defined the accuracy index to evaluate the performance of our method: the R-Squared (R^2). This ranges from 0 to 100%, where the best fit is that closest to 100%.

Comparative tests were carried out with different parameters. Firstly, using the trajectory prediction model constructed in this paper, the data were divided into one-year training and three-year testing sets ($K = 1$) to predict the position information (longitude and latitude) of buoys in the future. The accuracy of the position and the longitude and latitude of the buoys were 82.90% and 84.75%, respectively, with a comprehensive accuracy of 83.82%. Then, this process was repeated for three-year training and one-year testing sets ($K = 3$). The accuracy of the prediction and the longitude and latitude of the buoy were 86.03% and 89.85%, respectively, with a comprehensive accuracy of 87.94%.

The geographical significance of the movement of the water cycle is that through the movement of the water cycle, water continuously moves and transforms, keeping the various water bodies of the Earth in a constantly updated state, thereby maintaining the dynamic balance of global water. The hydrological cycle has a profound and widespread impact on global geography. Due to the limitations of observational data, the understanding of the 3D Pacific circulation in the oceanographic community has long focused on surface circulation, while there is a significant knowledge gap regarding the structure, variability, and link to climate change of the flow in the deep subsurface mesosphere. The equatorial mesospheric current plays an important role in the redistribution of matter and energy over a distance of about one kilometer. It can transport the highly dissolved oxygen of the western and central Pacific to the low-oxygen region of the eastern Pacific, which is also important for biogeochemical research.

Ocean currents influence and constrain a wide range of physical, chemical, biological, and geological processes in the ocean, as well as the formation and change in climate and weather over the ocean. Understanding and mastering the laws of ocean currents, large-scale air–sea interactions, and long-term climate change are therefore of great importance for fisheries, shipping, pollution discharge, and military applications.

6. Summary and Perspectives

The objective of this study was to employ deep learning techniques to forecast the trajectory of Argo buoys, enabling target tracking and emergency support. Given the significant spatio-temporal variations and high vacancy error rates in oceanic big data, a series of preprocessing activities were undertaken to enhance the efficiency of data delivery for predictive modeling purposes. These activities include filling in the missing data, excluding anomalies, standardizing the data, and selecting the relevant features. Two traditional deep learning techniques were employed for buoy prediction by matching buoy patterns,

resulting in favorable prediction outcomes through well-defined prediction tasks. A deep learning network called SRU was developed specifically to predict the trajectory of Argo buoys, with its parallel structure ensuring a good predictive performance. To address issues such as overfitting and vanishing gradients, we incorporated a skip connection strategy and spatio-temporal attention TSFFAM mechanism into the SRU to expedite the convergence. Our attention model utilized the SRU deep learning network structure to successfully predict buoy trajectory information, while overcoming the challenges associated with a lengthy prediction time and difficult training commonly encountered with the traditional LSTM models. To evaluate both the effectiveness and efficiency of our method, we conducted comprehensive multi-model comparison experiments along with validation using real-world observations from diverse geographic areas. The key results are as follows:

- (1) In the multi-model comparison trial, the SRU using the TSFFAM model attained the highest efficacy and significantly surpassed the standard LSTM method. Moreover, the SRU approach is competent in the real-time prediction of Argo trajectory.
- (2) Upon reviewing factual observations from various geographical regions, the SRU with TSFFAM model yielded the most satisfactory results when contrasted with the existing methods. This was largely due to its implementation of the skip connection optimization strategy in conjunction with parallel computing. Despite it not being the fastest model evaluated in this study, the SRU with TSFFAM approach is adequately swift in the vast majority of applications.

The experimental results demonstrate that the buoy trajectory prediction models presented in this paper achieved a high level of accuracy, which is highly significant for the fields of maritime search and rescue, as well as target tracking. Furthermore, deploying the proposed prediction model for practical target tracking at the earliest opportunity can enhance the emergency support capabilities and broaden its application across various domains. During global climate change, adjustments made in predicting buoy trajectories not only impact the fundamental laws of physics, but also exceptional and irregular climate phenomena, such as mesoscale eddies and turbulence outcomes. This affects the sensitivity of nonlinear and chaotic systems and makes them difficult to track and predict using conventional linear models. Therefore, it is imperative to improve the forecasting algorithms' accuracy by incorporating nonlinear deep learning methodologies that are essential for enhancing the predictive capabilities and facilitating informed decision making about future events. In the future, our aim is to enhance the performance of the SRU framework by integrating more effective optimization and acceleration techniques to increase its long-term prediction capabilities. This study has demonstrated significant effects through attention mechanism utilization, which will be further expanded upon in future research endeavors.

To cope with global warming, humanity must also continue to deepen its exploration of the oceans. In the future, buoys will also be used to monitor carbon storage and possible temperature increases in the ocean at depths of more than 2000 m. Measuring chlorophyll will provide information on the amount of biological activity and, ultimately, more information on the concentration of carbon dioxide in the ocean and atmosphere. The optical sensor can determine the color of the ocean water to reflect the activity of microalgae at the bottom of the food chain and then help infer what has happened in a particular area of the ocean.

To gather this information, researchers are developing sensors to measure carbon, chlorophyll, acidity, and concentrations of nutrient such as nitrate and phosphorus in seawater, and even to collect genomic data.

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